

API Endpoint Diagnostic Tool

Turn endpoint evidence into a clear next diagnostic action.

Sharad Kashyap • PHP 8.2+ • Framework-independent

401

REACHABLE

AUTHENTICATION
REQUIRED

A status code is evidence - not an explanation

Teams often lose time between seeing a failure and knowing what to inspect next.

RAW SIGNAL

401

Unauthorized

QUESTIONS LEFT OPEN

Is the token missing?
Is the scheme wrong?
Are scopes insufficient?
Did a proxy return the response?

COST

Guesswork

Repeated checks
and slow handoffs

The missing layer is interpretation.

Raw response evidence becomes an actionable report

INPUT

GET /protected

HTTP 401
Content-Type: JSON
WWW-Authenticate: Bearer
Duration: 125 ms



DIAGNOSTIC OUTPUT

Request Status

FAILED

Likely cause

**Authentication credentials are
missing or invalid.**

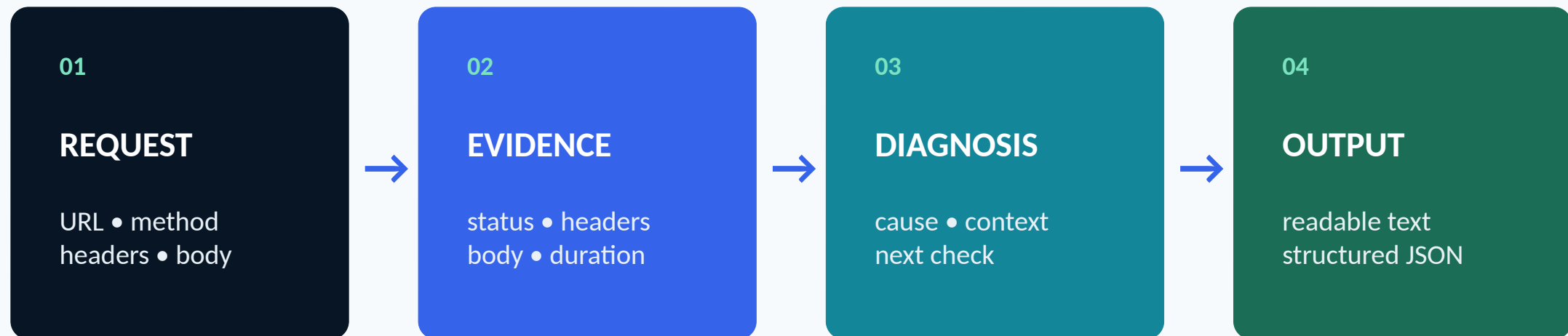
Suggested check

**Verify token, API key and
Authorization header format.**

Less guessing. Faster isolation. Cleaner client communication.

Four stages separate evidence from interpretation

Each layer has one job, making the system easier to test, extend and trust.



Framework-independent by design

The HTTP probe can evolve without rewriting diagnostic rules.

Diagnostic intelligence goes beyond a lookup table

Four evidence lenses work together before the report is produced.

TRANSPORT



DNS • timeout • SSL/TLS • refused connection

HTTP



Specific codes plus complete 2xx-5xx families

HEADERS



Allow • Retry-After • WWW-Authenticate

CONSISTENCY



Malformed JSON • HTML mismatch • invalid 204 body

Result: evidence-aware guidance without pretending to know server-side facts it cannot see.

Security controls are built into the diagnostic path

The tool limits what it can request and what it can reveal.

REQUEST BOUNDARY

HTTP and HTTPS only

- Redirect protocols restricted
- Redirect depth limited
- Request timeout enforced

OUTPUT BOUNDARY

Secrets stay out of reports

- Bearer credentials
- API keys and query tokens
- Passwords and URL credentials

The tool diagnoses authorized endpoints; it does not bypass access or modify remote services.

One diagnosis serves both people and automation

The same report can be read in a terminal or consumed as JSON.

HUMAN-READABLE

Request Status: Failed
Endpoint Reachable: Yes
HTTP Status: 429
Response Type: JSON

Likely Cause: Rate limit exceeded.
Suggested Check: Retry after 120 seconds.

MACHINE-READABLE

```
{  
  "request_successful": false,  
  "http_status": 429,  
  "response_type": "JSON",  
  "suggested_check":  
    "Retry after 120 seconds."  
}
```

Useful for debugging sessions, support handoffs and automated evidence capture.

Quality is enforced locally and on every GitHub push

The repository does not rely on a green demo alone - it carries repeatable engineering checks.

36

automated tests

91

assertions

L8

PHPStan analysis

4

PHP versions in CI

PHP 8.2

PHP 8.3

PHP 8.4

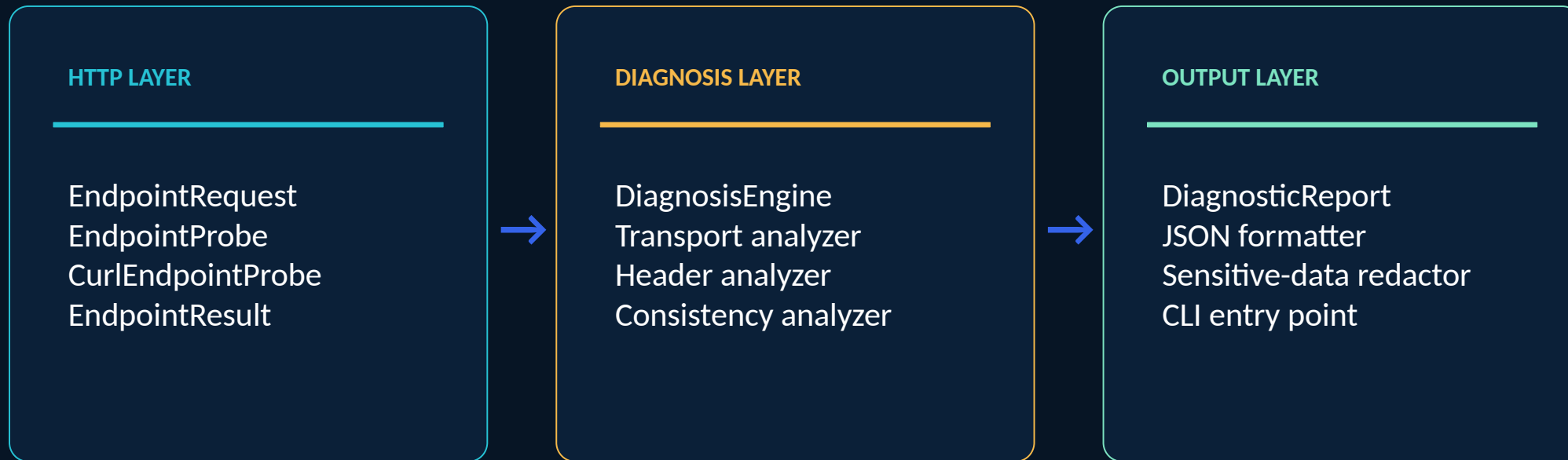
PHP 8.5

composer quality = PHPUnit + PHPStan

GitHub Actions status: passing

The architecture isolates change and keeps the core testable

Request execution, diagnostic reasoning and presentation remain separate concerns.



Benefit: new evidence rules or output formats can be added without rewriting the request engine.

Three practical workflows share one evidence model

Designed for small fixes today, with room for broader API tooling later.

- | | | |
|----|----------------------------|---|
| 01 | Freelance debugging | Turn a client's failing endpoint into a concise evidence-backed diagnosis. |
| 02 | Support handoff | Share likely cause, next check and structured context without exposing secrets. |
| 03 | Automation input | Feed JSON reports into logs, issue templates or future diagnostic workflows. |

Clear boundaries make the diagnosis more credible

The report distinguishes likely cause from guaranteed root cause.

WHAT IT DOES

- Captures endpoint evidence
- Classifies common failure patterns
- Recommends the next check
- Produces safe, reusable output

WHAT IT DOES NOT DO

- Bypass authentication
- Change remote configuration
- Guarantee unseen server-side causes
- Automatically repair the endpoint

Evidence → interpretation → recommended investigation

Ready for a real endpoint diagnostic scenario

Clone, install and run the tool against an authorized test endpoint.

```
git clone https://github.com/sharadkashyapdev/  
api-endpoint-diagnostic-tool.git
```

```
composer install  
php bin/diagnose.php <endpoint-url> --json
```

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tests

PHP 8.2-8.5
CI passing

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A focused tool. A clear diagnosis. A practical next
step.